Measurements and Units

Numbers and units:

Tera: 10^{12}

Giga: 10^{9} G

Mega: 10^{6} M

Kilo: 10^{3} K

---

deci: 10^{-1} d

centi: 10^{-2} c

milli: 10^{-3} m

micro: 10^{-6} μ

nano: 10^{-9} n

SI units:

1- Length : meters (m)

2- Time : seconds (s)

3- Mass : Kilograms (Kg)

4- Temperature : Kelvin (K)

5- Current : Ampere (A)

Derived quantities:

1- Area : m^2

2- Volume : m^3

3- Speed : m/s

4- Acceleration : m/s^2

5- Density : Kg/m^3

To find the volume of an object:

1) Regular Shapes: Formulas

2) Irregular shapes:

a) If the object sinks in water:

$\Delta V = V_2 - V_1$ → volume of water only

V_2 ↓ volume of water + object

b) If the object floats in water, we use a sinker

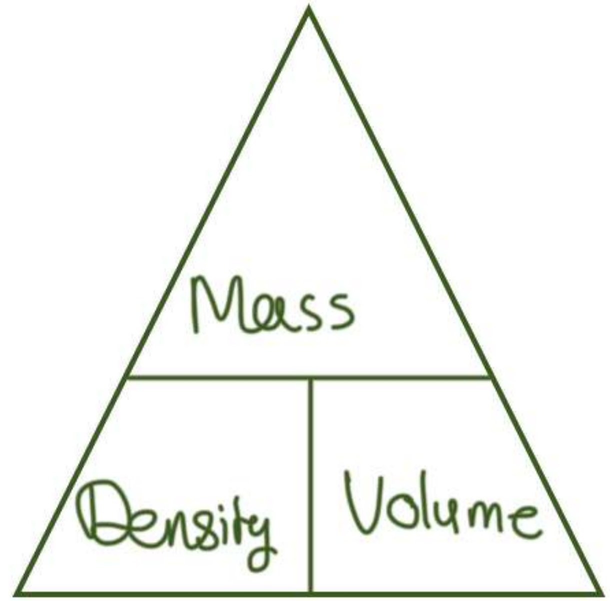
$\Delta V = V_3 - V_2$ → volume of water + sinker

V_3 ↓ volume of object + water + sinker

Density $= \dfrac{\text{mass}}{\text{Volume}} \rightarrow kg/m^3$

mass $=$ Density $\times$ Volume $\quad kg$

Volume $= \dfrac{\text{mass}}{\text{Density}} \quad m^3$



To convert:

a) from kg/m^3 to g/cm^3

$$\frac{kg}{m^3} \times \frac{1000}{100^3}$$

b) from g/cm^3 to kg/m^3

$$\frac{g}{cm^3} \div \frac{1000}{100^3} \rightarrow \frac{g}{cm^3} \times \frac{100^3}{1000}$$